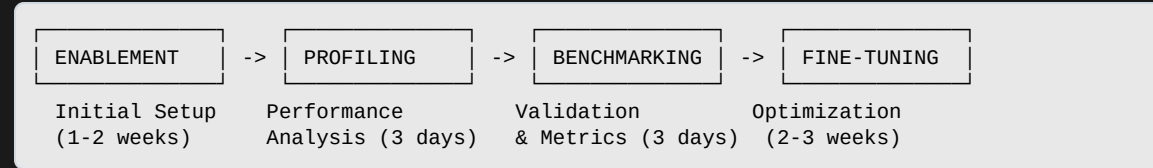


PAVS Model Lifecycle & Delivery Strategy 2026

4-Phase Model Lifecycle



Team Core Capabilities:

- **Model Enablement:** Off-the-shelf installation, dependency resolution, initial benchmarking
- **Performance Profiling:** Compute analysis, hotspot identification, bottleneck detection
- **Benchmarking & Validation:** Task metrics, Identify/suggest right model per domain/client, competitive analysis (GPU vs CPU vs NPU), CI/CD automation
- **Model Optimization:** Quantization (INT8, FP16), kernel optimization
- **Training Infrastructure:** Custom model training, domain-specific adaptation, fine-tuning on custom datasets
- **Deployment & Release:** Docker containerization, HuggingFace distribution, automated documentation

2026 Release Timeline

R1 2026 (Q1-Q3): Foundation

- Complete lifecycle framework for all models
- Streamline release logistics & CI/CD
- Deploy automated docs ([GitHub Pages](#))

R2 2026 (Q4-Q6): Scale

- Single SDK installer (Nvidia/Intel style)
- 8 production-ready models (P0/P1)
- Automated benchmarking pipeline
- Quantization & Infra Beta (vLLM, ONNX-RT, Llama.cpp, Lemonade)
- Sample apps (ChatBot, Object Detection, GStreamer+ROCm)

R3 2026 (Q7-Q9): Expand

- 8 more release-ready models
- Enhanced quantization & infrastructure
- Model deployment & robotics simulator
- Scale fine-tuning operations

R4 2026 (Q10-Q12): Production

- 2 additional production models (20 total)
- Multi-segment deployment (Robotics, Healthcare, Industrial)
- Model training infrastructure
- 2027 strategy planning

Open Source Release & Distribution Strategy

GitHub-Based Development & Release

Repository Structure:

```
physical-ai-sdk/  
├── models/  
│   ├── mobilesam/  
│   │   ├── gpu/           (Dockerfile, pyproject.toml, .venv)  
│   │   ├── npu_gpu/       (hybrid environment)  
│   │   └── npu/           (NPU-optimized)  
│   ├── yolov12/  
│   ├── whisper/  
│   └── [20 models...]  
├── benchmarks/           (automated CI/CD tests)  
├── docs/                 (auto-generated)  
└── tools/  
    ├── profilers/        (Lemonade, ROCm profilers)  
    └── optimizers/        (quantization, training)
```

CI/CD Automation Pipeline:

```
Developer Branch → Pull Request → Auto Tests  
    ↓              ↓              ↓  
Regression    Smoke Tests    Benchmarking  
    ↓              ↓              ↓  
Release Branch → QA Validation → GitHub Release  
    ↓              ↓              ↓  
Documentation Update    HuggingFace/tipp Push
```

Release Workflow:

```
Model Complete → Benchmarking → Optimization  
    ↓              ↓              ↓  
Package    Verify Metrics    Test Deploy  
    ↓              ↓              ↓  
GitHub Release → HuggingFace Push → Docs Update  
    ↓              ↓              ↓  
Community Access    Analytics & Feedback
```

HuggingFace Hub Distribution

What We Publish:

- 🧐 **Pre-optimized Model Weights:** ONNX, TensorRT, ROCm-ready
- 📦 **Quantized Variants:** INT8, FP16 for edge deployment
- 🐳 **Docker Images:** GPU/NPU/Hybrid ready-to-run containers
- 📊 **Benchmark Results:** Verified performance metrics included
- 📄 **Model Cards:** Documentation, usage examples, hardware requirements

HuggingFace Advantages:

- ✅ **Public Discovery:** Searchable by AI/robotics community
- ✅ **Version Control:** Track model iterations & improvements
- ✅ **Direct Download:** `pip install` or one-click deployment
- ✅ **Community Engagement:** Issues, feedback, contributions
- ✅ **Ecosystem Integration:** Works with Transformers, ONNX-RT, vLLM

Infrastructure Tools (Open Source Ready):

- **vLLM:** Efficient LLM serving & inference
- **Llama.cpp:** Optimized CPU/GPU inference
- **ONNX Runtime:** Cross-platform model deployment
- **Lemonade Profiler:** Performance analysis tool